‘Searchlight Scanner’ V5: Image
: Image-Recognition YOLO4 Training
Plan

1. Current (V2 Tactical Update)
Image
Object-Recognition Datasets

2. Previous (V2) Image – 9 Object
Object-Recognition Datasets

- To be included in the V5 Image Category

Category Name list

(Apparently actually from a COCO dataset Model:
tly actually from a COCO dataset Model: ssd_mobilenet_v2_fpnlite_320x320_coco17_tpu
ssd_mobilenet_v2_fpnlite_320x320_coco17_tpu-8 (1).tar.gz)

People [People]
Car [Car]
Bicycle [Bicycle]
Motorcycle [Motorcycle]
Airplane [Airplane]
Bus [Bus]
Train [Train]
Truck [Truck]
Boat [Boat]

Augmentation Augmentation
New Image Recognition Images V5 – Step 1:

12 Annotated Datasets Ready for

Image-Recognition Bounding-Box-Annotated (not
Annotated (not polygon) image datasets, ready for augmentation:
polygon) image datasets, ready for augmentation:

Note: Preferred Bounding box Format: YOLO4 (.txt
YOLO4 (.txt) files -for compatibility with AugmentImg, Mobilenet
Mobilenet and the
model.onnx files.

(Some earlier Bounding Box Annotations are currently in the Pascal VOC 1.1 (.x
(Some earlier Bounding Box Annotations are currently in the Pascal VOC 1.1 (.xml) format, which is not compatible, for
ml) format, which is not compatible, for
further annotation, with AugmentImg, where
where Total Desired Images = (Original (including background)
(including background) + Augmented).

Latest CVAT (and Roboflow) Annotated V5 - 12
12 Folders (for 25X AugmentImg) Aug>:

12 New Annotation Folder Names (Including YOLO Annotations and Original Images)
(Including YOLO Annotations and Original Images):

CVAT_AircraftCrashed_Images_YoloAnnotations_24June2025.zip
CVAT_AircraftCrashed_Images_YoloAnnotations_24June2025.zip

CVAT_PersonsOutdoors_Images_YoloAnnotations_24June2025.zip
CVAT_PersonsOutdoors_Images_YoloAnnotations_24June2025.zip

CVAT_PersonsThermalCloseUp_CommUse_Images
CVAT_PersonsThermalCloseUp_CommUse_Images_YoloAnno_UseThis24June2025.zip

CVAT_Powerline_Images_YoloAnnotations_24June2025.zip
CVAT_Powerline_Images_YoloAnnotations_24June2025.zip

CVAT_VehiclesIOffroad_Images_YoloAnnotations_24June2025.zip
CVAT_VehiclesIOffroad_Images_YoloAnnotations_24June2025.zip

CVAT_HelicopterCrashed_Images_YoloAnnotations_24June2025.zip
CVAT_HelicopterCrashed_Images_YoloAnnotations_24June2025.zip

CVAT_OceanDebris_Images_YoloAnnotations_24June2025
CVAT_OceanDebris_Images_YoloAnnotations_24June2025.zip

SnowTracks.Images_YoloAnnotations_24June2025.zip
SnowTracks.Images_YoloAnnotations_24June2025.zip

Ship2_ImageYolAnno_24June2025.zip

CVAT_OilSpill_Images_YoloAnnotations_24June2025.zip
CVAT_OilSpill_Images_YoloAnnotations_24June2025.zip

CVAT_Shipwake_Images_YoloAnnotations_24June2025.zip
CVAT_Shipwake_Images_YoloAnnotations_24June2025.zip

CVAT_Vessels_Images_YoloAnnotations_24June2025.zip
CVAT_Vessels_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_AircraftCrashed_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_PersonsOutdoors_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_PersonsThermalCloseUp_CommUse_Images_YoloAnno_UseThis24June2025.zip https://sartechnology.ca/sartechnology/CVAT_Powerline_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_VehiclesIOffroad_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_HelicopterCrashed_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_OceanDebris_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/SnowTracks.Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/Ship2_ImageYolAnno_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_OilSpill_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_Shipwake_Images_YoloAnnotations_24June2025.zip
https://sartechnology.ca/sartechnology/CVAT_Vessels_Images_YoloAnnotations_24June2025.zip

Aug CVAT_Aircraft_Crashed_YoloAnnotations > (txt) [Aircraft Crashed] > Aircraft Crashed annotations Pascal VOC 1.1 XML format (by Nick) [Aircraft Crashed] Image Source: Aircraft_Crashed_ObjectDetection_01May2025.zip 253 images 260,710 KB Image Source: AircraftCrashed_19Jan2024.zip 2296 images 1,612,283 KB AugmentImg – Aircraft Crashed: 25X Target Total Desired Images (Original + Augmentations): 63725

Aug> CVAT_PersonsOutdoors_YoloAnnotations (txt) [People Outdoors] > Persons annotations Pascal VOC 1.1 XML format (by Nick) [People Outdoors] Image Source: PersonInNature__12May2023.zip 1977 images 1,465,041 KB
Image Source: PersonsInNature_Object_30Oct2024.zip 20 images 27,922 KB
AugmentImg – People Outdoors 25X Target Total Desired Images (Original + Augmentations): 49925

Aug> CVAT_PersonsThermalCloseUp_CommUse_Yolo1p1_AnnotationFiles (txt) [Persons Thermal] Task607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_20June2025 (txt) [Persons Thermal Image Source: PersonsThermalCloseUp_20May2024.zip 2056 images 74,410 KB
AugmentImg – PersonsThermalCloseup 25X Target Total Desired Images (Original + Augmentations): 51400

Aug> CVAT_Powerline_YoloAnnotations (txt) [Powerlines]
> Powerline annotations Pascal VOC 1.1 XML format (by Nick) [Powerlines]
Image Source: PowerlinesCommUse_18June2025.zip 1001 images 1,337,526 KB
AugmentImg – Powerlines 25X Target Total Desired Images (Original + Augmentations): 25025

Aug> CVAT_VehiclesOffRoad_YoloAnnotations (txt) [Vehicles Offroad] > Vehicle annotations Pascal VOC 1.1 XML format (by Nick) [Vehicles Offroad] Image Source: Vehicles_19Jan2024.zip 1051 images 899,574 KB Image Source: Vehicle_Object_30Oct2024.zip 5 images 9,142 KB AugmentImg – Vehicles Offroad 25X Target Total Desired Images (Original + Augmentations): 26400

Aug> Task1394363_HelicoptersCrashedAnno_Yolo1p1_20June2025 (txt) [Helicopter Crashed] > Helicopters_Crashed_Project27211_Job2478556_PASCAL_VOC1p1_1954Frames.zip (xml) 2,810 KB [Helicopter Crashed] HelicopterCrashed_Project272110_Pascal.zip 1,833 KB [Helicopter Crashed] Image Source: Helicopter_Crashed_19Jan2024.zip 1520 images 898,908 KB Image Source: Helicopter_Crashed_Additional27Images_02April2024.zip 27 images 20,171 KB Image Source: HelicopterCrashed_ObjectDetection_28April2025.zip 398 images 380,604 KB AugmentImg – Helicopters Crashed 25X Target Total Desired Images (Original + Augmentations): 48625

Aug> CVAT__OceanDebris_Task1463727_Yolo1p1_Annotations22June2025 (txt) [Ocean Debris]
> OceanDebris_Job2490543_PASCAL_VOC1p1_254Frames.zip (xml) 266 KB [Ocean Debris]
Image Source: OceanDebris_19Jan2024.zip 26 images 9,850KB
Image Source: OceanDebris_Object_30Oct2024.zip 34 images 260,710 KB
Image Source: OceanDebris_ObjectDetection_06April2025.zip 250 images 288,729 KB
Image Source: Ocean_ShipwakeNegative_19Jan2024.zip 340,757 KB
AugmentImg – Ocean Debris 25X Target Total Desired Images (Original + Augmentations): 13150

SnowTracks_ObjectDetection
Aug> SnowTracks.v1i_Annotated._Yolov4pytorch_16April2025 (txt) [Snow Tracks]
SnowTracks.v1i_AnnotatedAugmented._Yolov4pytorch_16April2025.zip (3X Augments) (.txt) 93,814 KB [Snow Tracks] Source: SnowTracks.v1i_Annotated._Yolov4pytorch_16April2025
Image Source: SnowTracks_ObjectDetection_06April2025.zip 500 images 642,623 KB
AugmentImg – SnowTracks 25X Target Total Desired Images (Original + Augmentations): 12500

Ship2
Aug> ship2_dataset_Annotated_from_roboflow_cleaned.v2-download_for_v4.1.yolov4pytorch_01April2025 (txt) [Ship] Ship2_v4i_AnnotatedAugmented_Yolov4pytorch_16April2025.zip (3X Augments) (.txt) 90,514 KB [Ship]
Image Source:
ship2_dataset_Annotated_from_roboflow_cleaned.v2-download_for_v4.1.yolov4pytorch_01April2025\train (train: 1993 images)
AugmentImg – Ship2 25X Target Total Desired Images (Original + Augmentations): 49825

Aug> CVAT_OilSpill_Yolo1p1_Annotations21June2025 (txt) [Oil Spill]

Image Source: OilSpill_19Jan2024.zip 136 images 59,919 KB
Image Source: Ocean_ShipwakeNegative_19Jan2024.zip 340,757 KB
AugmentImg – OilSpill 25X Target Total Desired Images (Original + Augmentations): 8800

Aug> CVAT_Shipwake_Yolo1p1_Annotations21June2025 (txt) [Shipwake]
Image Source: ShipWake__13May2023.zip 110 images 70,016 KB
Image Source: Ocean_ShipwakeNegative_19Jan2024.zip 340,757 KB
AugmentImg – Shipwake 25X Target Total Desired Images (Original + Augmentations): 8150

Aug> CVAT_Vessels_Yolo1p1_Annotations22June2025 (txt) [Vessels]
Image Source: Vessel_01April2025.zip 195 images 253,625 KB
Image Source: Ocean_ShipwakeNegative_19Jan2024.zip 40,757 KB
AugmentImg – Vessel 25X Target Total Desired Images (Original + Augmentations): 10275

Total 14,712 images Average: 732 images per category

'Searchlight Scanner' V5 - Total 21 Image Datasets - Training Procedure for YOLO4 Models

Included Total 21 (9 Original + 12 New) Image Datasets for V5 YOLO4 Training:

1. Previous V2 - 9 Original Image COCO Datasets:
[People] [Car] [Bicycle] [Motorcycle] [Airplane] [Bus] [Train] [Truck] [Boat]

2. Current (V2 Tactical Update) Image Object-Recognition - 3 Datasets:
[Powerlines] [Persons Thermal] [People]

3. Additional (V3) Image Object-Recognition Images - 8 Datasets
(including images retrained from Classification to Object Detection):
[Aircraft Crashed] [Helicopter Crashed] [Vehicles Offroad] [People Outdoors]
[Shipwake] [Vessel] [Oil Spill] [Ocean Debris]

4. New (V4) Image Object-Recognition Images - 2 Datasets
[Ship] [Snow Tracks]

Training Step 2. Bounding Box Annotation
(Only for any remaining un-annotated image collections):

Draw tight bounding boxes around image objects using GITHUB CVAT (Computer Vision Annotation Tools) https://www.cvat.ai/ then export the annotation files in Yolo V 1.1 (txt) format for these image collections. This Yolo V1.1 exported format creates a companion annotation .txt file for each annotated image.

Step 1. For an existing CVAT Project.... Create a new (annotation) Task.
Step 2. Upload images to the new Task.... Submit and Open (images).
Step 3. Jobs... Annotation (i.e. draw bounding boxes) tightly around the objects.
Step 4. Requests... Export Annotations in (Yolo Y 1.1 format)... Finished... then click the three (unlabelled) dots … to begin the actual download of the Annotation Project (zip) File.

Training Step 3. Image Augmenting – using AugmentImg

(i.e. creating multiple slightly-different copies of each original image)

Image Augmenting - using AugmentImg:

Step 3 - AugmentImg - Free Image Augmentation Procedure

https://github.com/zamalali/AugmentImg

Step 2 AugmentImg - Free Image Augmentation Procedure (also supports YOLO (txt) and COCO (json) annotation formats) https://github.com/zamalali/AugmentImg

AugmentImg is a user-friendly image augmentation tool for computer vision tasks. It offers a no-code graphical interface to apply various augmentations to images and their annotations. Simply input the desired count and select the types of augmentations. It supports both YOLO (txt) and COCO (json) annotation formats.

This program is non-proprietory, fast, free to use, and can rapidly create a very large number of augmented images, ready for model training.

Note:

Background 'Null' images should also be included with the bounding-box annotated target images for augmentation.

AugmentImg Installation:

Run the following commands in your terminal

```
pip install augmentimg
augment-img
```

Ensure you have Python 3.6+ installed (from the Microsoft store) on the Windows computer, then clone the repository and install dependencies:

```
git clone https://github.com/zamalali/AugmentImg.git
cd AugmentImg
pip install -r requirements.txt
```

To run AugmentImg on Windows, Python 3.11 must first be downloaded and then installed, from the Microsoft Store. The Windows CMD command prompt is then used to run pip to install Augmentimg.

Once installed AugmentImg can be run from directly from the Windows desktop using a right-click, create-shortcut link to e.g. C:\Users\Martin\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local packages\Python311\Scripts\augment-img.exe

AugmentImg has the following 12 available transformation parameters, with fixed default values, that can be selected for augmentation:

Rotate 90: (custom-changed internally for STI to Rotate +/-15 degrees)
Flip Horizontally: Left or Right
Flip Vertically: Up and Down
Brightness: +/- 20%
Blur: 0.5 px

Saturation (Hue): 0.5 Randomly selected
Contrast: 0.5 ?
Sharpness: 0.5
Gamma (guass:) 0.5
CLAHE 0.5 (localized enhanced contrast within the image)
HSV: (Hue, Saturation and Brightness Value)
Shift, Scale, Rotate: Fully Random
(Total Desired Images: main.py edited to increase the maximum from 1000 to 1000000 desired

images) AugmentImg Procedure:

1. Select Image Folder: Initially, users must select the image folder containing the original images they wish to augment. Note: Images with bounding boxes and also images without bounding boxes, i.e. 'Null' background images, should both be included for augmentation.

2. Select Annotations (file-path)
* Paste the full (not partial) folder-filepath to the main folder containing the individual YOLO V 1.1 annotation text files (not list) e.g. to OilSpill001.txt, Oilspill002.txt, Oilspill003.txt etc. e.g.

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\Annotated_SplitImageFoldersForAugmentationTest \CVAT _OilSpill_Task1461701_Yolo1p1_Annotations21June2025\obj_train_data

Functioning_CVAT_Augmented_Filepaths:

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24 June2025\ CVAT_HelicopterCrashed_AnnotationFiles\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24 June2025\ CVAT_AircraftCrashed_YoloAnnotations\obj_Train_data
C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24 June2025\ CVAT_HelicopterCrashed_Images\1

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24 June2025\ CVAT_SnowTracksImages_Yolo1p1_Annotations_06July2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\Persons_ThermalCloseUp_CommUse\CVAT_Pers onsThermalC loseUp_CommUse_Yolo1p1_AnnotationFiles\Task607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_2 0June2025\ obj_train_data

Note:
(i) While all of the original CVT_PersonsThermalCloseup #1 to #25 subfolders with the image filename type of: e.g. IMG_0017-2_jpg.rf.d8ac36b0e5cf84be4e6486dc00d5088c.jpg - correctly created new augmented files from these numbered original image subfolders,

(ii) files of the filename type:
WIN_20240315_17_27_47_Pro_mp4-0019_jpg.rf.e2ba8deef9ba4a306ce76655b67b3c52.jpg displayed a DOS command-screen 'Annotation file does not exit' warning message and did not create any augmented images into the subsequent empty 'augmentations' subfolders,

- despite these 'WIN_ ...' filenames also being present within the same annotation obj_train_data subfolder and the same train.txt file as the 'IMG_...' image files.

(iii) Summary: Approx 1,000 'IMG_ jpg' original files correctly created 25X augmented images into the #1 to #25 image subfolders, while the remaining 1,000-image #26 to #52 image subfolders, for the 'WIN_' format image files, therefore remain without any augmentation images.

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\T ask607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_20June2025_1\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Task607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_20June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Task607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_20June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Task607617Persons_ThermalCloseUp_CommUseYolo1p1Anno_20June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_PersonsOutdoors_YoloAnnotations\obj_Train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_PersonsOutdoors_YoloAnnotations\obj_Train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June202 5\CVAT_Powerline_YoloAnnotations\obj_Train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_VehiclesOffRoad_YoloAnnotations\obj_Train_data
C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_VehiclesOffRoad_YoloAnnotations\obj_Train_data\obj_Train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_VehiclesOffRoad_YoloAnnotations\obj_Train_data\obj_Train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_OceanDebris_Task1463727_Yolo1p1_Annotations22June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\
CVAT_OceanDebris_Images_YoloAnnotations_24June2025.zip\CVAT__OceanDebris_Task1463727_Yolo1p1_Annotations22J une2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Shipwake_Task1462812_Yolo1p1_Annotations21June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Shipwake_Task1462812_Yolo1p1_Annotations21June2025\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_Vessels_Task1462870_Yolo1p1_Annotations22June2025\obj_train_data

2. Choose Transformations: Users can then go through the ("Select Annotations") checklist and select all 12 transformations, by clicking on their corresponding checkboxes.

3. Specify Total Desired Images: Users should enter the "Total Desired Images" Count i.e. (Original + 25X Augmented images), max 1,000,000 (originally was 999), in the provided input field.

4. Augment Images: By clicking the "Augment" button, the application will apply the selected transformations to the original images and save the augmented images-only, along with each augmented-image's new annotation .txt file.
C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_HelicopterCrashed_AnnotationFiles\obj_train_data

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_HelicopterCrashed_Images\1

C:\Downloads\VizeIt_ImageRecognition\ImageClassifications\AnnotatedImageCollections_ForAugmentation_24June2025\ CVAT_SnowTracksImages_Yolo1p1_Annotations_06July2025\obj_train_dataannotation .txt file, into the automatically created 'augmented' sub-folder, continuing to generate augmented images until the Total (Original + Augmented), Desired Image Count is reached.
(Note: For YOLO V 1.1 an individual annotation .txt file is also created for each new augmented image).

Recommended 'Total Desired Images': 25X (Original + 25X Augmented) total number of images. 25X Augmentation Rate: Approx 560 original images/hour.

Note: Due to AugmentImg originally having a maximum of 999 Total Desired Images (Original + Augmented) for 25X augmentation…
(i) First use the Windows 'Folder Splitter' program to create numbered image-subfolders from just the original image folder, each containing just 40 original images per sub-folder.
(ii) AugmentImg can then select this image sub-folder and create a total of (max) 999 images = 40 Original Images + 959 augmented images per subfolder i.e for a 25X augmentation.
Note: To increase the limit of 999 total augmented images to any number you want to, you only need to change your main.py line number 55 to any number you like to have.

```
 self.spinBoxDesiredImages.setRange(1, 1000)
```

Edited to: self.spinBoxDesiredImages.setRange(1, 1000000)

Just change this 1000 to any number that you like to have as your max limit. Make sure you don't change anything else. Then rerun the GUI. Note: There must be no image-subfolders below the single top-level image folder.

Note: BOTH the Original AND its Augmented images must be subsequently trained together by the image-recognition model-creation model.onnx routine.

Note: AugmentIng supports image augmentation for images with YOLO (.txt) and COC-annotations.txt and_classes.txtO (.json) annotation file-formats.
- including Yolo 1.1 with an obj_train_data folder (for 'Select Annotations') and obj.data, obj.names and train.txt files. - and Yolo4 Pytorch with Test, Train & Valid folders, each containing _annotations.txt and _classes.txt files.

The YOLO4 .txt (bounding box) annotation file-format is compatible with YOLO4, which works with Mobilenet and the generated model.onnx file format.

(Pascal VOC Version 1.1 annotation files have a XML annotation file-format).

Recommended 'Total Desired Images': 25X original number of images.

'Total Desired Images' Example:
If you're doing around 15 augmentations per image for a folder with 2,000 originals, you'll end up with roughly 32,000 total images. That's a solid (15 Augmentations + 1 Original) = 16X scale-up.
- For general image classification tasks, something like 10X to 20X scale-up is usually more than enough. - For trickier datasets like outdoors, changing lighting, or smaller original sets, going up to 20X or even 25X can help, like you did with the "Persons Outdoors" set.
- Beyond that, the extra images often don't add much value, and it just ends up increasing training time. - Somewhere between 15X to 20X is a sweet spot for most cases.
- The 23X you tried sounds great for more complex datasets, but for clean indoor or structured sets, even 10X might be fine.

Alternate Augmentation Information: Roboflow Default Augmentation Parameters:

(Alternative Recommended Augmentation Values taken from roboflow)
Rotation: Rotate the image by a specified degree. Between -15 degrees and +15 degrees Creates 2 copies Flip: Flip the image horizontally or vertically. use Horizontal and Vertical Creates 2 copies Blur: Apply a Gaussian blur to the image. Up to 0.3px (random Gaussian blur) Creates 1 copy Saturation: Adjust the image's color saturation. Between -25% and +25% Creates 2 copies Contrast: Modify the image's contrast level. Between 0.4 to 1.8 for outdoor images Creates 2 copies Sharpness: Enhance or reduce the sharpness of the image. Between 0.5 to 2.0 for drone images Creates 2 copies Random Crop: Crop random parts of the image. 0% Minimum Zoom, 20% Maximum Zoom Creates 2 copies Random Erase: Randomly erase parts of the image. - Creates 1 copy
Total Number of Augmented Copies (per original image): 14 total augmented copies
Generate Total Number of Augmentation Copies created: 25X (Total of Original + Augmented Images.)

Affine Transformations:
- Apply affine transformations like scaling, translations, and rotations

These additional AugmentImg 23X augmented images should really help to increase the robustness of the object detection during subsequent model training.

11 Annotated & 25X Augmented Yolo V1.1 Image Recognition Datasets - Ready for AI Model Training

https://sartechnology.ca/sartechnology/CVAT_Vessels_Images_Augmented_07July2025_OriginalAndAugmented_16July2025.zip https://sartechnology.ca/sartechnology/CVAT_Vessels_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_VehiclesIOffroad_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_VehiclesIOffroad_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_SnowTracks_AugmentedObjectDetection_Images_OriginalAndAugmented

_16July 2025.zip
https://sartechnology.ca/sartechnology/CVAT_SnowTracksImages_Yolo1p1_Annotations_06July2025.zip

https://sartechnology.ca/sartechnology/CVAT_Shipwake_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_Shipwake_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_Powerline_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_Powerline_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_PersonsThermalClosep_Images_OriginalAndAug_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_PersonsThermalCloseUp_CommUse_Images_YoloAnno_UseThis24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_PersonsOutdoors_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_PersonsOutdoors_Images_YoloAnnotations_24June2025.zip

https://sartechnology.ca/sartechnology/CVAT_OilSpill_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_OilSpill_Task1461701_Yolo1p1_Annotations21June2025.zip

https://sartechnology.ca/sartechnology/CVAT_OceanDebris_Images_OriginalAndAugmented_16July2025.zip
https://sartechnology.ca/sartechnology/CVAT_OceanDebris_Task1463727_Yolo1p1_Annotations22June2025.zip

https://sartechnology.ca/sartechnology/CVAT_HelicopterCrashed_Images_OriginalAndAugmented_16July2025.zip https://sartechnology.ca/sartechnology/CVAT_HelicopterCrashed_AnnotationFiles.zip

https://sartechnology.ca/sartechnology/CVAT_AircraftCrashed_Images_OriginalAndAugmented_16July2025.zip https://sartechnology.ca/sartechnology/CVAT_AircraftCrashed_YoloAnnotations.zip

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------------ Image Training Step 4. Model Training Using MobileNet SSD and Tensor Flow:

- Image Recognition Training for the 'Searchlight Scanner V5' - Background Information

Image Recognition training for the Searchlight Scanner is performed on all the augmented, bounding-box annotated Image Datasets. (Bounding boxes are used instead of the larger polygons to provide faster, less-computational, edge processing). MobileNet SSD is the required image recognition inferencing model and TensorFlow is used to run this inferencing on the scanner.

4.1. SSD-Mobilenet-v1 requires bounding box annotations, instead of polygons, to reduce the computational demand for edge-optimized image-recognition models.

4.2. Training the model on a local GPU device, instead of Google Colab, was more reliable, once the Python script had been optimized. The Google Colab (free tier) generated inconvenient pauses and interrupts.

It is therefore recommended that s local GPU device be used to train all the new image-recognition categories into the new model.

4.3. The (Version 5) (26 April 2024) (Persons Outdoors – only) image-recognition model created is called ssd-mobilenet.onnx

4.4 Detailed Image Model Training Plan Procedure:

Model Training Step 1.
Use pretrained YOLOv3 model - look into other versions but this one doesn't have the open source requirements that some of the more recent models have.
Install PyTorch and repository dependencies : pip install torch torchvision -r requirements.txt

Model Training Step 2.
Organize dataset into dataset/images/{train,val,test} and dataset/labels/{train,val,test} The split can be done via a (python) script, similar to the one I shared, below.
A 70/20/10 split is a decent option I think but this can be adjusted, depending on the size of the data set.

Python Augmentations scripts:
The script augment.py - takes two directories containing images and annotations and produces a specified number of copies with various transformations applied.

The script split.py - splits the images and annotations into test, training, and validation subsets and generates the corresponding text files.

The scrip to create_tf_records.py - creates the record files needed for training.
augment.py
create_tf_records.py
split.py

Model Training Step 3.
Create data.yaml from CVAT's obj.names.
This defines the dataset structure, specifying class names, where the images folders are located, etc. -
Validate annotations by visualizing bounding boxes to ensure normalized coordinates and correct class IDs.
The class names in the yolo txt files have to match data.yaml and coordinates should be in the range [0,1]
ex. <class_id> 0.5 0.5 0.15625 0.104167

Model Training Step 4.
Train with pre-trained yolov3.pt weights command:
python train.py --data data.yaml --cfg yolov3.yaml --weights yolov3.pt --epochs 100 --img-size 640
Evaluate on val and test sets command : python test.py --data data.yaml --weights best.pt

Warnings:
- Check the model's license to ensure no open source requirements.
- Don't skip annotation validation, since CVAT errors like mismatched class IDs can ruin training.
- Check data.yaml paths are absolute, not relative since this may cause an issue.

I think this will be a good starting point for the group.
They'll naturally have to do some research of their own but with multiple people working on it I'm sure they'll get a good grasp of it.

Training Results Metrics:
Mean Average Precision (mAP)
Combining precision and recall can tell us at a glance the overall general performance of our model, and serves as a good metric for relative performance when compared to other models.
A model with 90% mAP generally performs better than a model with 60% mAP, but it's possible that the precision or recall of the 60% mAP model provides the best solution for your problem. Smaller models will almost always make less accurate predictions. The metric we use to evaluate object detection accuracy is called mean average precision, a measure of how well the predicted bounding boxes overlap with ground truth bound boxes. The main takeaway of these results is that if your models performance is saturated (greater than 95% mAP) you will not see much of an accuracy tradeoff.

Precision = True Positive / (True Positive + False Positive)
Let's say we build an apple detection model and give it 1000 apples to detect.
If it successfully detects 500 apples (TP), incorrectly detects 300 apples (FP), and fails to detect 200 apples (FN): We'd have a Precision of 500/800, or 62.5%

Recall = True Positive / (True Positive + False Negative)
Using the same apple example, our model would have a Recall of 500/700, or 71%.
"e.g. Initial work should therefore be focused on creating a high recall minimum viable product (MVP) with an emphasis placed on reducing the rate of False negative detections, as a false negative represents a missed leak".

False Negative (FN): The object was in the image and the model did not draw a box, "there was something there but the model missed it." This occurs most often when the object is seen at a weird angle, lighting, or state not captured in the model's training data. It can be reduced by increasing object representation in the training data.
We can reduce the FP rate by understanding what the model falsely identifies as an apple and adding that to the training set. If the green pears look like the green apples, label the pears!

Need to improve recall? Reduce the False Negative rate by adding (Augmenting) more variations of apples at different angles and levels of proximity to the camera.

Version Training Results:
Personsthermal_ drone/1 v1
Your PersonsThermal_Drone (2024-07-17 6:30pm) (Defaults) model finished training in an hour and achieved 92.4% mAP, 95.0% precision, and 89.7% recall.
Train 349 Valid 99 Test 50 = 498 Total images
Model Type: Roboflow 3.0 Object Detection (Fast) Checkpoint: COCO

(Roboflow) Version Training Results:
Personsthermal_drone/2

Your PersonsThermal_Drone (2024-07-17 7:41pm) (Augmented) model finished training in 24 minutes and achieved 92.0% mAP, 95.1% precision, and 89.9% recall.
Train 1047 Valid 99 Test 50 = 1196 Total images
Model Type: Roboflow 3.0 Object Detection (Fast) Checkpoint: personthermamal_drone/1

Preprocessing Step: Auto-Orient: Applied
Resize: Stretch to 640x640

Roboflow Augmentations Step: Outputs per training example: 3 (i.e. 3X)
Flip: Horizontal, Vertical
Crop: 0% minimum Zoom, 20% Maximum Zoom
Rotation: Between -15 degrees and + 15 degrees
Shear: +/1 10 degrees Horizontal, +/- 10 degrees Vertical
Blur: Up to 0.3px (random Gaussian blur)
Brightness: Between -15% and + 15%
Saturation: Between -25% and + 25%
Hue: Between -15 degrees and + 15 degrees
